Improving Low-Resource Grammatical Error Correction with Synthetic Data Across Typologically Diverse Languages

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Abstract

Grammatical Error Correction (GEC) remains a challenge for low-resource languages due to limited gold data. Without parallel corpora of learner errors with corrections by native speakers, GEC models cannot be reliably evaluated. Using data from LangCorrect, a community-driven language-learning platform where L1 speakers correct L2 learners' essays, we provide benchmarks for two low-resource languages: Turkish and Vietnamese. Specifically, we evaluate mBART-50, fine-tuned on synthetic data generated using the SeLex-RT method recently applied to Russian and Ukrainian by Gomez and Rozovskaya (2025). Thus, our work has three primary contributions: first, we publish two parallel corpora of human-corrected sentences in Turkish and Vietnamese; second, we provide benchmarks for these two low-resource languages that lack prior work on GEC; and third, we investigate whether SeLex-RT improves scores for typologically diverse languages beyond the closely related Slavic languages already examined. Our results indicate that SeLex-RT is a viable method for generating better synthetic data for Turkish and Vietnamese, suggesting that the technique can assist GEC models on a wide range of languages.

1 Introduction

Grammatical Error Correction (GEC) has seen substantial progress in high-resource languages, particularly English, driven by large annotated corpora and advances in neural sequence-to-sequence models (Bryant et al., 2023); however, many languages considered high-resource in the context of NLP more broadly are still low-resource for the topic of GEC because they lack the gold data necessary for evaluation: learner text with errors and corrections. Without this data as a reference, there is no way to determine how close to human standards a GEC model performs.

While lack of gold data is a non-starter for GEC modeling, lack of training data presents another problem. Fine-tuning language models with parallel error-ful and error-less text has been repeatedly shown to improve the scores of GEC models. (e.g., Lichtarge et al., 2018; Náplava and Straka, 2019; Gomez and Rozovskaya, 2025) Unlike evaluation data, however, there is no reason that training data must be produced by humans. Thus, a promising direction in GEC has been the use of synthetic data—monolingual data, corrupted with artificial errors—for model training and fine-tuning.

Various approaches have been proposed to generate artificial grammatical errors, including rule-based and neural methods (Grundkiewicz et al., 2019; Stahlberg and Kumar, 2021), and ranging from the character level (e.g. Kiyono et al., 2019) to the morpheme level (e.g. Flachs et al., 2021) to the lexical level. SeLex-RT (Gomez and Rozovskaya, 2025) is a recent method that generates injects lexical errors using round-trip machine translation, and has demonstrated improvements for Russian and Ukrainian; however, due to the similarity of these two languages, the question of whether SeLex-RT can generalize across typological diversity remains open.

We target this question, and fill the gap of GEC research on non-Indo-European languages, using two typologically diverse examples: Turkish (Turkic, agglutinative) and Vietnamese (Austroasiatic, analytic). Each presents distinct challenges for error modeling and correction.

- Create new and publicly available gold GEC datasets for Turkish and Vietnamese, using learner essays and native speaker corrections on LangCorrect, a free community-driven language learning platform.
- Provide benchmarks for GEC models in Turkish and Vietnamese, neither of which have been explored.

- Examine the cross-linguistic adaptability of SeLex-RT as a method of creating synthetic data for fine-tuning GEC models.

2 Related Work

2.1 Approaches

Since most prior work in GEC has focused on Indo-European languages, limited attention has been given to languages like Turkish and Vietnamese. (Marier et al., 2025) Here we discuss a handful of representative work for both of these languages.

Work on Vietnamese GEC has been limited in scope, largely due to lack of annotated data. Nguyen et al. (2019) use a deep learning approach to correcting spelling errors in Vietnamese. Their work advances research by focusing on realistic spelling mistakes, which often create new words, as opposed to mistyped words. However, the dataset they use for evaluation, a parallel corpus of Vietnamese news text with and without spelling errors injected, is synthetic, and thus, not quite a gold standard. Do et al. (2021) address the same problem, and do use a human-produced dataset for evaluation. They build a Transformer based model trained on a large dataset injected with synthetic errors, but evaluate it with human-assembled data of spelling errors and corrections. While both of these works advance Vietnamese GEC, spelling errors are only a subset of the type of grammatical errors that learners produce, and the work neglects morphological, lexical, and fluency errors.

GEC research on Turkish is slightly more advanced. Kara et al. (2023) released GECTurk, a corpus of 130,000 parallel sentences with synthetic errors injected using writing rules informed by Turkish Language Association. They use the GECTurk corpus across three task formulations: neural MT, sequence tagging, and a pretrained decoder-only model. While they demonstrate good results, the evaluation sets are primarily synthetic, and the only gold dataset is a very small (albeit human-produced) set of ~150 human movie reviews.

Uz and Eryiğit (2023) solved an important problem in low-resource GEC by creating ERRANT-TR, a GEC evaluation method based on the English-specific tool ERRANT. (Bryant et al., 2017) This tool is important to accurately evaluate Turkish GEC, as the language's agglutinative nature makes tasks like tokenization and morphological analysis very different from English.


```
{
  "post_id": 173,
  "sentence_id": 1506,
  "learner_level": "A1",
  "original_sentence": "Bana göre çok
      iyidir!",
  "corrected_sentence": "Bana göre bu
      çok iyi!",
  "correction_types": [],
  "note": null
}
```


Figure 1: An example annotation from the raw Turkish dataset.

2.2 Dataset

We introduce new gold datasets for Turkish and Vietnamese GEC, derived from LangCorrect, a platform where native speakers correct learner-produced text. We extract 4,075 sentence-level aligned pairs for Turkish from 273 posts and 1,411 for Vietnamese from 101 posts. The average words per sentence and the number of corrected sentences per sentence can be found in Table 1.

To ensure quality, we asked one native speaker for each language to review 10% of the entries and confirm that the corrections are accurate, that is, addressing all grammatical errors, not introducing any new errors, and improving fluency where necessary. Each annotator found that over 90% of the sentences in their sample met these standards. The resulting datasets consist of learner-generated errors paired with corrected sentences. We will release these datasets publicly to support future research.

3 Methodology

3.1 Data Processing

The raw dataset sent to us by LangCorrect contained entries with the following attributes: post id, sentence id, learner level, original sentence, corrected sentence, correction types, and note. Unfortunately, the correction types were not available during data scraping, but more than half of the entries contain notes in which the corrector has either left a comment on the error type, explained the cultural context, or simply stated that they are unsure of the meaning of the original sentence. For this project, we worked only with the original and corrected sentences from the data, but we believe other aspects of the dataset could be useful in future work.

Additional filtering was conducted to ensure

Table 1: Summary statistics of the Turkish and Vietnamese GEC datasets.

Language	Original Sentences	Posts	Avg. Words/Sentence	Corrected Sentences/Original
Turkish	2,029	273	9.74	2.03
Vietnamese	650	101	13.30	2.15

dataset quality. We looked at every single entry for Turkish and Vietnamese and:

- Removed every entry that had the same original and corrected sentence with a note that specified that the corrector was unsure of what the learner was trying to imply.
- Removed a few entries whose corrected sentences included text in a language beyond Turkish/Vietnamese.
- Aligned original sentences that had the sentence number in the original sentence but not in the corrected sentence. For example: *original: 6. Son aylarında... kızıyordular. corrected: Son aylarda... kızıyorlar.*

The cleaned dataset consisted of 4,075 examples in Turkish and 1,411 examples in Vietnamese. was then formatted for input to the mBART-50 model for training. We ran a script and converted the dataset into the {"src": . . . , "tgt": . . . } format, as shown in Figure 2 below.

{"src": "Khi nào trời dừng mưa?", "tgt": "Khi nào trời ngừng mưa?"}
{"src": "Có lẽ ngày kia không mưa nhưng mặt trời cũng không chiếu sáng.", "tgt": "Có lẽ ngày kia sẽ không mưa nhưng mặt trời cũng sẽ không chiếu sáng."}
{"src": "Chúng tôi đi công viên đi dạo.", "tgt": "Chúng tôi đi công viên đi dạo."}
{"src": "Thư trương của nhà máy này làm việc nhiều hơn công nhân.", "tgt": "giám đốc của nhà máy này làm việc nhiều hơn công nhân."}
{"src": "Hãy để tôi trong hòa bình vì tôi mệt lắm!", "tgt": "Hãy để tôi yên vì tôi mệt lắm!"}
{"src": "Nếu ngực của bạn bị đau, hãy tức thì gọi bác sĩ.", "tgt": "Nếu ngực của bạn bị đau, hãy lập tức gọi bác sĩ."}
{"src": "Vì sao bạn không thích cắt hành tây?", "tgt": "Vì sao bạn không thích cắt hành tây?"}
{"src": "Tương lai tôi định đi Việt Nam để nói tiếng Việt với người Việt.", "tgt": "Tương lai tôi định đến Việt Nam để nói tiếng Việt với người Việt."}

Figure 2: Example of the formatted Vietnamese dataset from LangCorrect.

Since the gold dataset was not large, we split the data to be 50% training and 50% testing.

3.2 Model

We use mBART-50 (Liu et al., 2020), a multilingual sequence-to-sequence model pretrained on large-scale multilingual corpora. mBART-50 is well-suited to low-resource grammatical error correction because it is a multilingual sequence-to-sequence model pretrained on denoising and multilingual translation objectives. Since GEC can be framed as translating "incorrect language" into "correct language," mBART-50's encoder-decoder architecture naturally fits the task. Its multilingual pretraining enables transfer learning from high-resource, related languages, which is especially useful when the target language has little annotated GEC data. We fine-tune separate models for each language.

3.3 Synthetic Data Generation

Synthetic data generation is the process of creating artificial training examples that imitate real data. In the context of low-resource grammatical error correction (GEC), synthetic data is especially useful because there are often not enough naturally occurring error–correction sentence pairs available for training. However, synthetic data must be generated carefully. If the artificial errors are too random or unrealistic, the model may learn patterns that do not match real learner mistakes.

To address this problem, synthetic data can be generated by taking grammatically correct sentences and automatically introducing realistic errors into them. These artificial errors may include spelling mistakes, incorrect verb forms, missing articles, wrong word order, agreement errors, or punctuation mistakes. The original correct sentence is then used as the target output, while the corrupted sentence becomes the input. This creates artificial error–correction pairs that can be used to train or fine-tune a GEC model.

We generate synthetic training data using Selex-RT (Gomez and Rozovskaya, 2025). Synthetic errors are injected into clean monolingual corpora drawn from the Turkish Wikipedia Corpus (Linguistic Data Consortium, 2015) and the Leipzig Mixed Vietnamese Corpus (Goldhahn et al., 2021). To support round-trip translation within Selex-RT, we use OPUS-MT models (Tiedemann and Thottingal, 2020) for the following language pairs: English↔Vietnamese, English↔Turkish. 25 back-translated sentences were produced for each of 100k sentences randomly samples from each of these corpora, ultimately producing confusion sets for lexical items from 2.5M translations, for each

language.

3.4 Training Setup

For each language, we train three models:

- **Gold-only:** mBART-50 fine-tuned on available gold data.
- **Synthetic Baseline + gold:** mBART-50 fine-tuned on available gold data, followed by data corrupted with character perturbations.
- **Synthetic + gold:** mBART-50 fine-tuned on available gold data, followed by data corrupted with lexical replacements (using SeLex-RT) as well as character perturbations.

3.5 Evaluation

So that others can reliably compare scores with our benchmarks, MaxMatch scorer (Dahlmeier and Ng, 2012) was originally selected as our evaluation metric; however, the technique was ultimately not a good fit for our dataset, as many of the erroneous sentences in our datasets had multiple references. MaxMatch scorer penalizes corrections that are valid but don't match any single reference closely.

We instead employ ERRANT for our evaluation technique. (Bryant et al., 2017) While ERRANT scores have been used in multilingual contexts, error type detection and analysis cannot be extended to languages beyond English. Thus, we only report an $F_{0.5}$ outcome, and no information on the types of errrors that were produced, and fixed.

4 Experiments

4.1 Experimental Conditions

For each language, we compare the three training conditions: (1) fine-tuning on gold data alone, (2) fine-tuning on gold data and a synthetic baseline (character perturbations) and (3) fine-tuning on gold data and synthetic data generated by SeLex-RT and character perturbations. This will help us understand whether synthetic data improves GEC performance in low-resource settings beyond what can be achieved with limited gold supervision alone.

5 Results

Results are provided in Table 2. For each language, the worst results were attained by fine-tuning mBART-50 on gold data alone, and the best results were attained by fine-tuning on both gold data and data generated using lexical and character-level errors. For both models, precision scores were consistently higher than recall scores. SeLex-RT proves to provide the best result consistently for both languages.

System	Prec	Rec	$F_{0.5}$
Turkish			
Gold Data Only	0.1863	0.0401	**0.1078**
Gold + Char (Synthetic Baseline)	0.2555	0.0549	**0.1476**
Gold + SeLex-RT + Char	0.2710	0.0745	**0.1774**
Vietnamese			
Gold Data Only	0.0825	0.0377	**0.0667**
Gold + Char (Synthetic Baseline)	0.0956	0.0344	**0.0705**
Gold + SeLex-RT + Char	0.1115	0.0457	**0.0866**

Table 2: ERRANT Results on Turkish and Vietnamese. Prec = precision, Rec = recall.

6 Discussion

The generally low benchmark scores for Vietnamese and Turkish reveal useful information for the advancement of GEC in these languages. For starters, mBART-50 may not be the best multilingual Transformer to fine-tune with gold and/or synthetic data. It would be worthwhile to explore other options that accommodate Turkish or Vietnamese, as the baseline here is not very high.

As expected, fine-tuning the model with gold data alone provided the lowest scores. Fine-tuning with additional synthetic data improved the scores of the model, and most importantly, fine-tuning with synthetic data generated by SeLex-RT produced the best scores of all three techniques. Given that this technique had previously been applied only to Russian and Ukrainian, these results demonstrate that the methodology can be reliably applied to typologically diverse languages and still improve scores. In this way, our research corroborates the versatility of SeLex-RT as a way to improve GEC models in a (very) low-resource settings, even for agglutinative and analytic languages like Turkish and Vietnamese.

Nevertheless, we think that creating benchmarks for both of these languages by using a gold dataset can be very beneficial in the work that is to come.

6.1 Future Work

We have a number of plans to improve upon the work we have thus conducted to provide GEC benchmarks on Vietnamese and Turkish.

First, we seek native speakers of Turkish and

Vietnamese to review the entirety of our gold data to clean it more precisely. Such annotators could classify errors, verify corrections, and eliminate corrected sentences that do not improve fluency. Increasing the size of these datasets would also be beneficial to ensure a robust evaluation metric.

Additionally, we plan to employ ERRANT-TR (Uz and Eryiğit, 2023) as our Turkish evaluation metric. Due to time constraints, we opted to use the traditional ERRANT scorer, but a metric catered to Turkish is far more reliable. Unfortunately, no equivalent exists for Vietnamese, so using ERRANT will likely be our best path forward.

We also hope to improve both the size and the quality of the datasets used for fine-tuning. Although synthetic data is useful for low-resource grammatical error correction, the effectiveness of the model depends heavily on how realistic and diverse the generated examples are. Therefore, expanding the dataset with more sentence pairs that reflect natural grammatical errors would likely improve the model's ability to generalize to real-world input. We would tackle this by not only using character perturbations as a baseline, but spelling errors, morphological errors, etc.

Finally, we would like to conduct longer and more extensive training. Due to limited time and computational resources, our model was trained on only a subset of the synthetic data available to us. With more time, we could fine-tune the model on a larger portion of the dataset and experiment with different training settings, such as learning rates, batch sizes, and numbers of epochs.

7 Conclusion

To summarize our work, we have created gold datasets for two languages considered low-resource in the realm of GEC: Turkish and Vietnamese. We fine-tuned mBART-50 on this gold data, alongside data corrupted with synthetic errors, and evaluate the model's quality using ERRANT. Our results demonstrate that synthetic data improves our model's ability to correct grammatical errors, particularly when that data includes lexical errors injected using the SeLex-RT methodology. This suggests that SeLex-RT can be reliably applied to typologically distinct languages, beyond the closely-related Russian and Ukrainian. Our model's scores provide benchmarks for GEC in two low-resource languages that have suffered from limited attention and lack of data. Moving forward, we plan to advance this research by tackling each of the points listed in the Future Work section.

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